Lehrstuhl für Software und Systems Engineering
TUM School of Computation, Information and Technology
Technical University of Munich

Agentic AI for Code Generation

Seminar Report — SS26

Thua Duc Nguyen

Supervisor: Derui Zhu
TUM School of Computation, Information and Technology, Technical University of Munich

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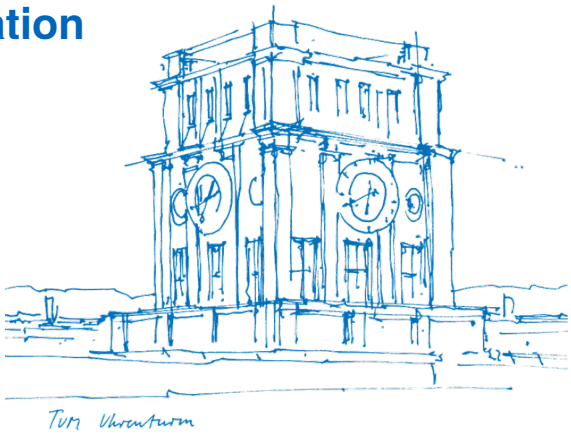
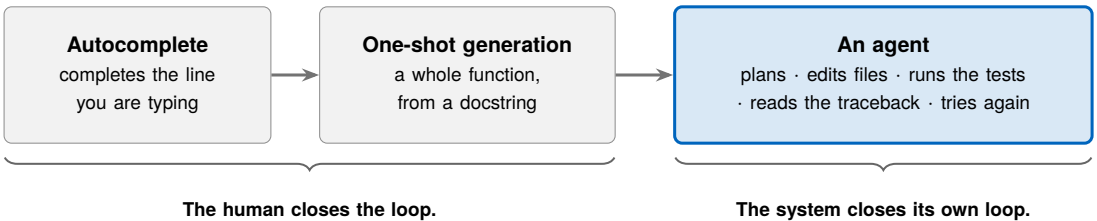
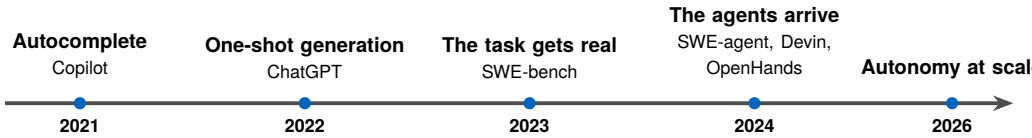


What is agentic AI for code generation?



Plain language in — a code change out.

Three years, three eras

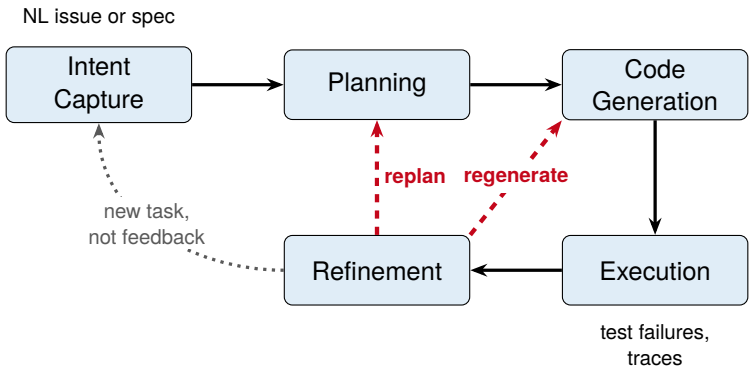


under 2 % ⟶ **over 70 %**

of real GitHub issues resolved autonomously [10, 21]

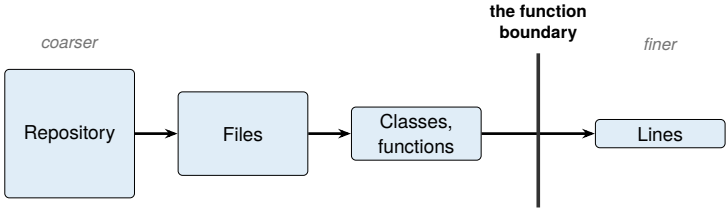
(two different rulers — defined in a few slides)

The agent pipeline



Five stages — **one vocabulary for every system in this talk.**

Planning is localisation, not design

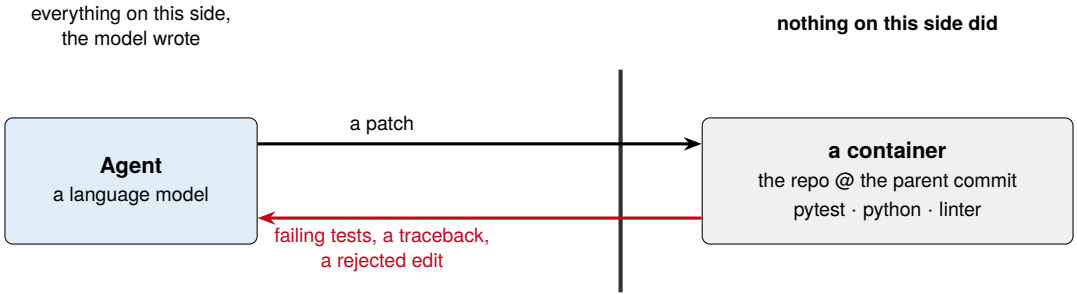


Above it, localisation *is* the task — ranking functions *helps* [26].

Below it, it hurts — *no* line pointer beats *perfect* localisation [24].

Reasoning about *where* the code is — not *how* to write it.

Execution: the code meets the environment



Not a score — *a symptom*, still to be read.

Refinement: what to do with the signal

- **Repair from the trace** [9]
- **Explain your own code** [3] — *only where tests exist*
- **Sample N, rerank with a trained critic** [16]

Agents improve their own code.

What SWE-bench measures

One instance [10] = a real merged PR: closed an issue + touched the tests.

The agent sees the repo before the merge + the issue; **returns** a patch.

Graded by running the tests — the bug fixed, nothing broken.

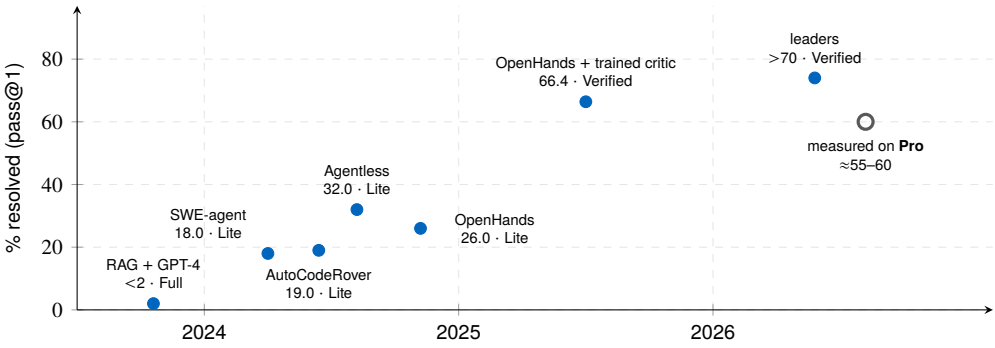
Resolve rate = both, on the first patch.
Every number from here on is this one.

One benchmark, four splits:

- **Full** — 2,294, every scraped PR
- **Lite** — 300, single-file patches
- **Verified** — 500, human-screened
- **Pro** — 1,865, held-out repos

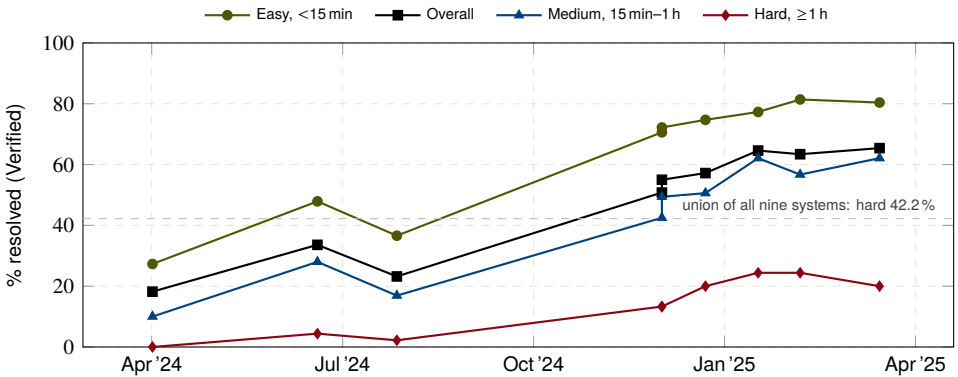
Rates are not comparable across splits. Verified is the most *curated* split, not the hardest.

Three years, in numbers



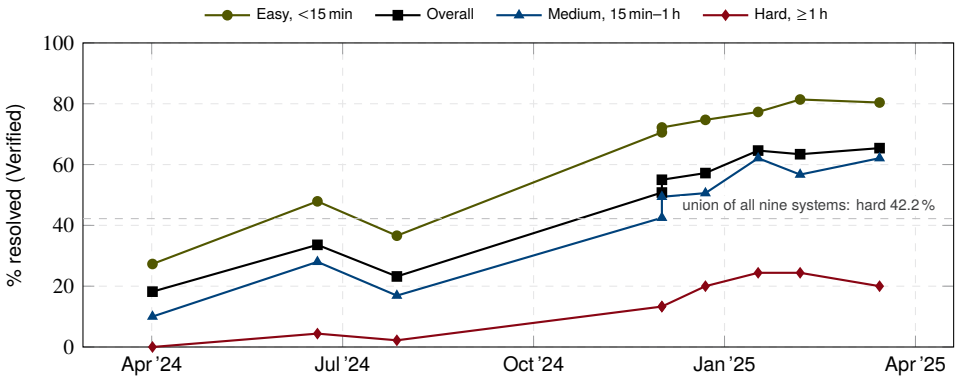
Splits are **not comparable** — read the *climb*, not the deltas. [10, 25, 26, 23, 22, 16, 21, 17]

What the climb hides



Nine systems with published per-tier results; tiers = annotator-estimated fix time ($n = 194/261/45$ of 500) [6, 14].

What the climb hides



Nine systems with published per-tier results; tiers = annotator-estimated fix time ($n = 194/261/45$ of 500) [6, 14].

A system can gain ten points — and solve no additional hard task.

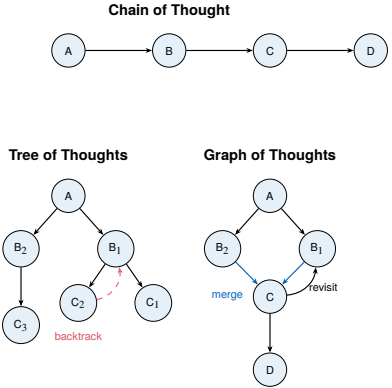
Conclusion

- In three years: under 2 % → **over 70 %**.
- **What changed is the loop** — the system runs its own code, and reads what comes back.
- The climb was mostly the easy strata — **hard tasks stalled near 20 %** [6]. *The climb is not over.*

Backup: the numbers behind the stage walk

- **Planning, above the function boundary** [26]. AutoCodeRover adds spectrum-based fault localisation to rank *functions*: $+3$ **points** on SWE-bench Lite.
- **Planning, below it** [24]. D4C hands the model the failing test and the *whole function*, and gives it *no line-level pointer at all* — beating systems given perfect localisation by **10 %**. Below the function boundary, a pointer fights the model's infilling prior.
- **Refinement, with an explanation** [3]. Self-debug: up to $+12\,\%$ — but only where a test suite exists to run against.
- **Refinement, with a trained critic** [16]. OpenHands on Verified, best-of-N reranked: 60.6 → **66.4 %**. (On the deck: the numbers timeline.)

Backup: how are plans shaped?



The evaluator, not the branching, makes the structure useful — and it is what a repository-scale agent finds hardest to supply. *What is a half-finished patch worth, before any test has run?* SWE-Search [1] builds one (MCTS + LLM value function, $+23\,\%$ relative) — and must invoke a model at every node.

Backup: how do agents refine?

Mechanism	Closed by	Signal	Characteristic failure
Self-Refine [12]	**self**	Model critiques own output	Plateaus; nothing can contradict the generator
Self-debug, no tests [3]	**self**	Self-generated explanation	Explains the bug as intended behaviour
Self-debug, with tests [3]	**oracle**+**self**	Test verdict + model diagnosis	Bounded by test coverage
Reflexion [18]	**oracle** / **proxy**	Task reward where one exists; on HumanEval, self-generated tests	Filed as external, but its code result rests on a proxy
SelfEvolve [9]	**oracle**	Interpreter errors, traces	Silent (non-crashing) semantic bugs
D4C [24]	**oracle**	Failing tests + full function	Plausible-but-incorrect patches
CRITIC [8]	**oracle**	Self-critique routed *through* an external tool	Needs a tool covering the error class
Agentless validate [23]	**oracle**+**proxy**	Regression tests + generated reproduction tests	The proxy half inherits the model's misreading
Best-of-N + critic [16]	**oracle**+**proxy**	Executed rollouts ranked by a trained critic	Critic is a distilled proxy; cost scales with N

oracle = a real checker — it runs the code. **proxy** = a made-up checker — trained or generated. **self** = the model judging itself.

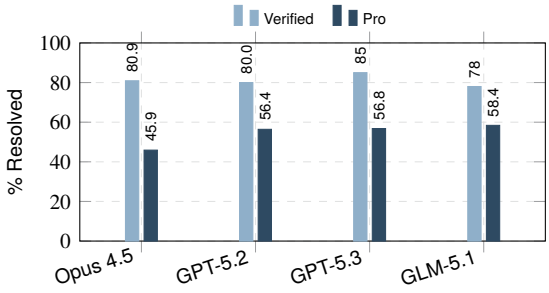
Backup: does the scaffold matter?

	SWE-agent custom ACI	**OpenHands** code-as-action	**mini-SWE-agent** ~100 lines of Python	**spread**
GPT-5-mini	38.8	38.6	**51.8**	13.2
Qwen3 Coder 480B	36.6	**54.0**	40.2	17.4
DeepSeek R1	**22.6**	18.0	11.0	11.6

Resolve rate (pass@1) on SWE-bench Verified, April 2026 [19, 20]. ACI = agent–computer interface: the command set the model is handed.

- Same model, same benchmark: the scaffold alone is worth **11–17 points**.
- **No scaffold wins twice** — the best scaffold *depends on the model*.

Backup: is the 70 % real?



The *same weights* drop ≈ 20–35 points on the contamination-resistant SWE-bench Pro [4].

Read only the floor. Pro is harder *as well as* fresher, and the column mixes harnesses — so the drop *size* is not a measurement of memorisation.

91 % of Verified is work a developer finishes *within the hour* [6]. A system can gain points while solving no additional hard task.

July 2026: the ruler itself is under audit. OpenAI flags ∼30 % of Pro's public tasks as broken and retracts its February recommendation to use it [15]. Quote *measured* Pro numbers, with that caveat attached.

Backup: what does running an agent cost?

- **Longer windows do not dissolve the bottleneck.** Retaining full *dependency* context works best; padding with loosely related code *degrades* correctness [13].
- **Context explosion and semantic drift.** Appending every observation buries the critical early information [11].
- **Cost is quadratic in turns.** The whole prefix is re-sent each turn, so tokens grow as $O(n^2)$ [7]. A property of the protocol, not an empirical finding; prompt caching cuts the constant, not the growth rate.
- **Resolve rate decouples from compute** [5]. Agentless leads on resolve rate (48 % with Qwen3-32B) at 83.1 API calls and 209.4 s per issue. The frontier is accuracy *per dollar*.
- **And verification is the human bottleneck.** In an RCT, experienced developers were 19 % *slower* with early-2025 AI tools — while believing they had been 20 % faster [2].

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